

Paper 1B practice 2

The block that came out too hot

IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **21 marks**

Show your working.

A student measures the specific heat capacity of aluminium. An electrical immersion heater and a thermometer sit in holes drilled in a solid aluminium block. The heater is switched on for exactly five minutes, and the ammeter and voltmeter readings hold steady throughout. The thermometer has 1 °C divisions.

Quantity	Reading	Uncertainty
mass of block, m	1.000 kg	± 0.001 kg
potential difference, V	11.8 V	± 0.1 V
current, I	3.85 A	± 0.05 A
heating time, t	300 s	± 1 s
temperature before heating	19.4 °C	± 0.5 °C
temperature after heating	33.6 °C	± 0.5 °C

The student assumes that all the electrical energy supplied warms the block, so that $VIt = mc\Delta T$.

- (a) State the temperature rise ΔT . Explain why its uncertainty is ± 1.0 °C and not ± 0.5 °C. [3]
- (b) Calculate the electrical energy supplied to the heater. [2]
- (c) Determine the specific heat capacity of aluminium from these readings, and state its unit. [3]
- (d) Calculate the percentage uncertainty in each measured quantity, and hence the percentage uncertainty in your value of c . [4]
- (e) Express c with its absolute uncertainty. [2]
- (f) The accepted value for aluminium is $900 \text{ J kg}^{-1} \text{ K}^{-1}$. Discuss whether this experiment is consistent with it. [2]
- (g) In practice some thermal energy escapes to the surroundings during the five minutes. State whether this makes the experimental value of c too high or too low, and explain your reasoning. [3]
- (h) Use your answer to (d) to identify the single change that would most reduce the uncertainty, and justify why it beats the alternatives. [2]

Answers

- (a) $\Delta T = 33.6 - 19.4 = 14.2^\circ\text{C}$. The rise is a *difference* of two separate readings, and each one carries its own $\pm 0.5^\circ\text{C}$. For a sum or a difference it is the *absolute* uncertainties that add, never the percentages, so the uncertainty in the rise is $0.5 + 0.5 = 1.0^\circ\text{C}$. (Subtracting the readings does not subtract their uncertainties: taking a difference always makes the absolute uncertainty worse, not better.)
- (b) $E = VIt = 11.8 \times 3.85 \times 300 = 1.36 \times 10^4 \text{ J}$ (13 629 J).
- (c) Rearranging $VIt = mc \Delta T$,

$$c = \frac{VIt}{m \Delta T} = \frac{13\,629}{1.000 \times 14.2} = 960 \text{ J kg}^{-1}\text{K}^{-1}.$$

- (d) c is built entirely from products and a quotient, so the *percentage* uncertainties add:

Quantity	Percentage uncertainty	Share of the total
V	$0.1/11.8 = 0.85\%$	9%
I	$0.05/3.85 = 1.30\%$	13%
t	$1/300 = 0.33\%$	3%
m	$0.001/1.000 = 0.10\%$	1%
ΔT	$1.0/14.2 = 7.04\%$	73%
total	9.6%	

- (e) 9.6% of 960 is 92, so $c = (9.6 \pm 0.9) \times 10^2 \text{ J kg}^{-1}\text{K}^{-1}$, that is $960 \pm 90 \text{ J kg}^{-1}\text{K}^{-1}$. The uncertainty is quoted to one significant figure and the value is rounded to match its precision.
- (f) The result spans 870 to 1050 $\text{J kg}^{-1}\text{K}^{-1}$, and the accepted 900 lies inside that interval, so the experiment *is* consistent with the accepted value. It is worth noticing that the central value is about 7% high rather than scattered either side, which hints at a systematic effect rather than random error; part (g) identifies it.
- (g) **Too high.** The student assumes every joule of VIt ends up in the block, but some escapes to the surroundings, so the energy that actually warms the block is less than VIt . The block therefore rises by less than it would with no losses: here 14.2°C against the $13\,629/900 = 15.1^\circ\text{C}$ expected for perfect insulation. Since $c = VIt/(m \Delta T)$ and ΔT appears in the denominator, a smaller measured rise gives a *larger* value of c .
- (h) The temperature rise. It contributes 7.0% of the 9.6% total, very nearly three quarters of it, so it is the only term whose improvement matters much. Either measure ΔT more precisely, with a digital probe reading to 0.1°C , or make ΔT much larger by heating for longer, so that the fixed $\pm 1.0^\circ\text{C}$ becomes a smaller fraction of it. The alternatives are not worth chasing: even perfect voltage *and* current readings would only bring the total from 9.6% down to 7.5%.